Open and Accessible Quantum Computer Benchmarking

Alessandro Cosentino

March 24, 2026 • video call

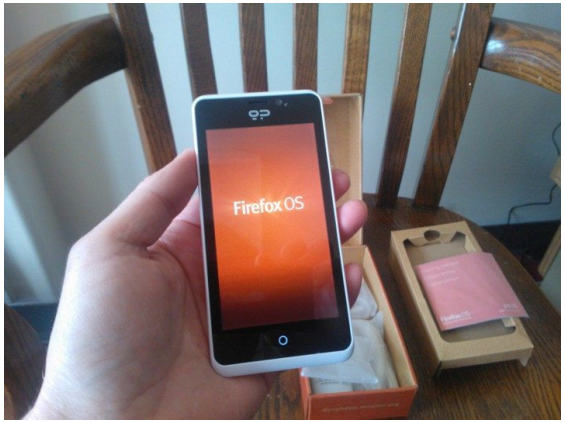
Mozilla Foundation | Unitary Foundation

Alessandro Cosentino

Metriq team · Unitary Foundation

Me and Mozilla...

- Firefox user for 15+ years
- Firefox OS user (more niche)
- Mozilla Outreachy mentor



Unitary Foundation

Because evolution is unitary.

Microgrants

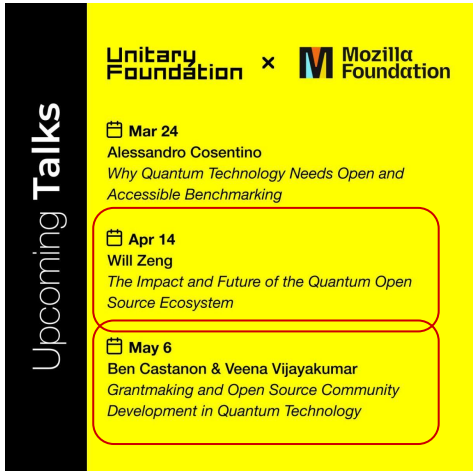
- We run a microgrant program to fund explorers across the world to work on quantum technologies. **100+ teams across 31 countries**.

Research

- in Benchmarking, Compilation, Error Mitigation

Community

- We host an open source quantum tech community **(4k+ open quantum tech developers)** that runs hackathons, community surveys and events.



LF

In classical computing,

we *can work* with proprietary benchmarks 🤷🏻

In quantum computing,

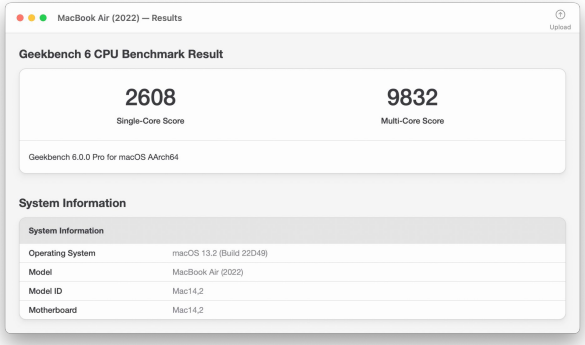
we need open benchmarks.

Classical technology benchmarking

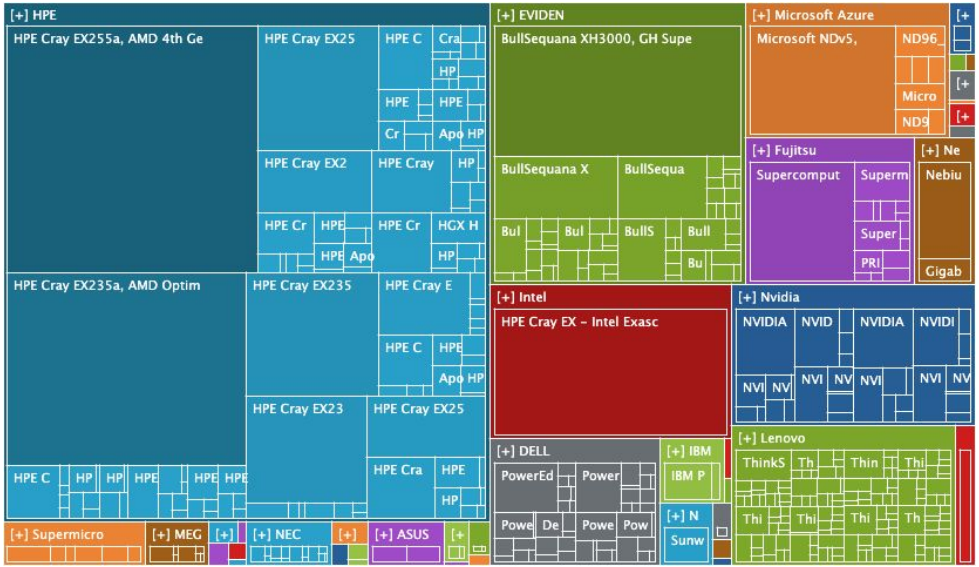
- ❑ Systems are comparable
- ❑ Easy access to systems
- ❑ Grounded in real-world workloads
- ❑ Constant cross-checking

👉 **Widely used and trusted benchmarks despite not always open!**

Examples (machines)



Geekbench · CPU / GPU · proprietary

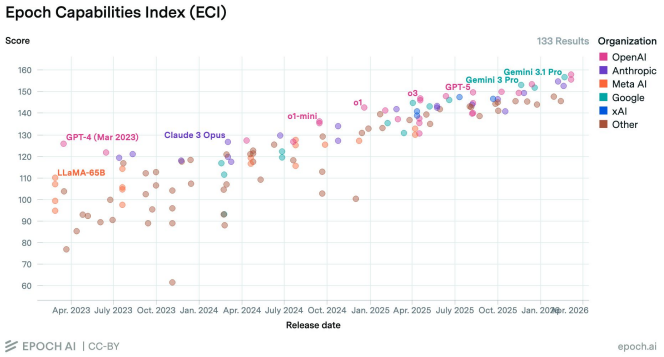


Linpack

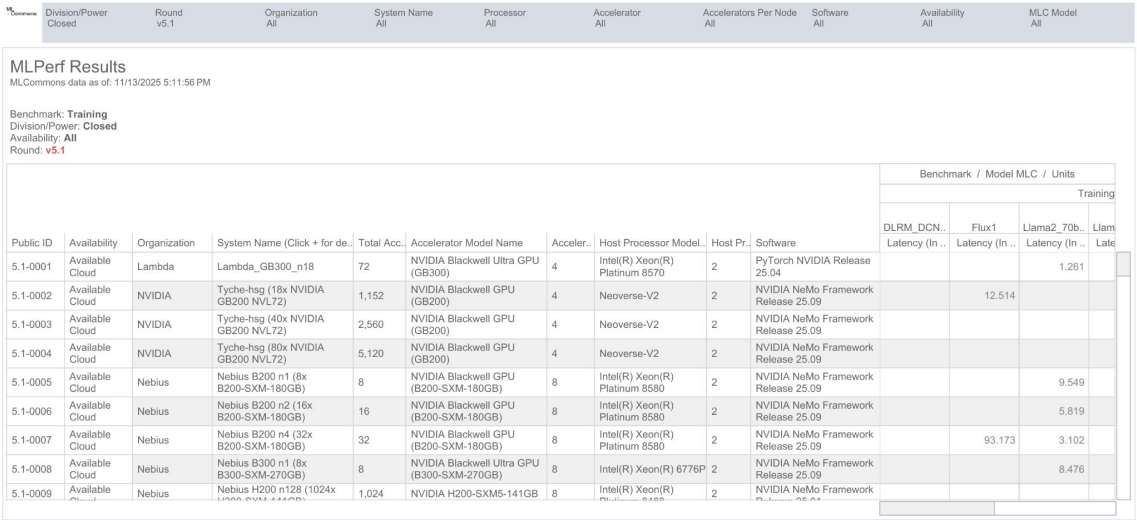
Top500 supercomputer

system-specific implementations

Examples (AI)



Epoch AI · models · meta-benchmarking



Division/Power	Round	Organization	System Name	Processor	Accelerator	Accelerators Per Node	Software	Availability	MLC Model
Closed	v5.1	All	All	All	All	All	All	All	All

MLPerf Results

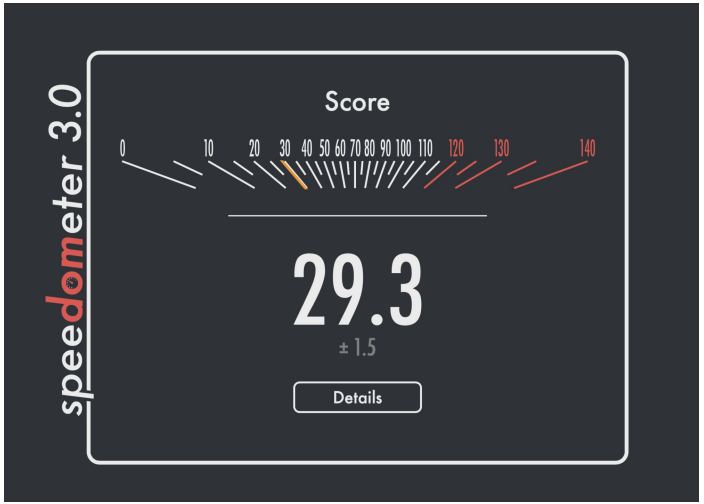
MLCommons data as of: 11/13/2025 5:11:56 PM

Benchmark: **Training**
Division/Power: **Closed**
Availability: **All**
Round: **v5.1**

										Benchmark / Model MLC / Units			
										Training			
										DLRM_DCN..	Flux1	Llama2_70b..	Llam
Public ID	Availability	Organization	System Name (Click + for de..	Total Acc..	Accelerator Model Name	Acceler..	Host Processor Model..	Host Pr..	Software	Latency (In ..	Latency (In ..	Latency (In ..	Late
5.1-0001	Available Cloud	Lambda	Lambda_GB300_n18	72	NVIDIA Blackwell Ultra GPU (GB300)	4	Intel(R) Xeon(R) Platinum 8570	2	PyTorch NVIDIA Release 25.04			1.261	
5.1-0002	Available Cloud	NVIDIA	Tyche-hsg (18x NVIDIA GB200 NVL72)	1,152	NVIDIA Blackwell GPU (GB200)	4	Neoverse-V2	2	NVIDIA NeMo Framework Release 25.09		12.514		
5.1-0003	Available Cloud	NVIDIA	Tyche-hsg (40x NVIDIA GB200 NVL72)	2,560	NVIDIA Blackwell GPU (GB200)	4	Neoverse-V2	2	NVIDIA NeMo Framework Release 25.09				
5.1-0004	Available Cloud	NVIDIA	Tyche-hsg (80x NVIDIA GB200 NVL72)	5,120	NVIDIA Blackwell GPU (GB200)	4	Neoverse-V2	2	NVIDIA NeMo Framework Release 25.09				
5.1-0005	Available Cloud	Nebius	Nebius B200 n1 (8x B200-SXM-180GB)	8	NVIDIA Blackwell GPU (B200-SXM-180GB)	8	Intel(R) Xeon(R) Platinum 8580	2	NVIDIA NeMo Framework Release 25.09			9.549	
5.1-0006	Available Cloud	Nebius	Nebius B200 n2 (16x B200-SXM-180GB)	16	NVIDIA Blackwell GPU (B200-SXM-180GB)	8	Intel(R) Xeon(R) Platinum 8580	2	NVIDIA NeMo Framework Release 25.09			5.819	
5.1-0007	Available Cloud	Nebius	Nebius B200 n4 (32x B200-SXM-180GB)	32	NVIDIA Blackwell GPU (B200-SXM-180GB)	8	Intel(R) Xeon(R) Platinum 8580	2	NVIDIA NeMo Framework Release 25.09		93.173	3.102	
5.1-0008	Available Cloud	Nebius	Nebius B300 n1 (8x B300-SXM-270GB)	8	NVIDIA Blackwell Ultra GPU (B300-SXM-270GB)	8	Intel(R) Xeon(R) 6776P	2	NVIDIA NeMo Framework Release 25.09			8.476	
5.1-0009	Available	Nebius	Nebius H200 n128 (1024x	1,024	NVIDIA H200-SXM5-141GB	8	Intel(R) Xeon(R)	2	NVIDIA NeMo Framework				

MLPerf · ML stacks
public specs

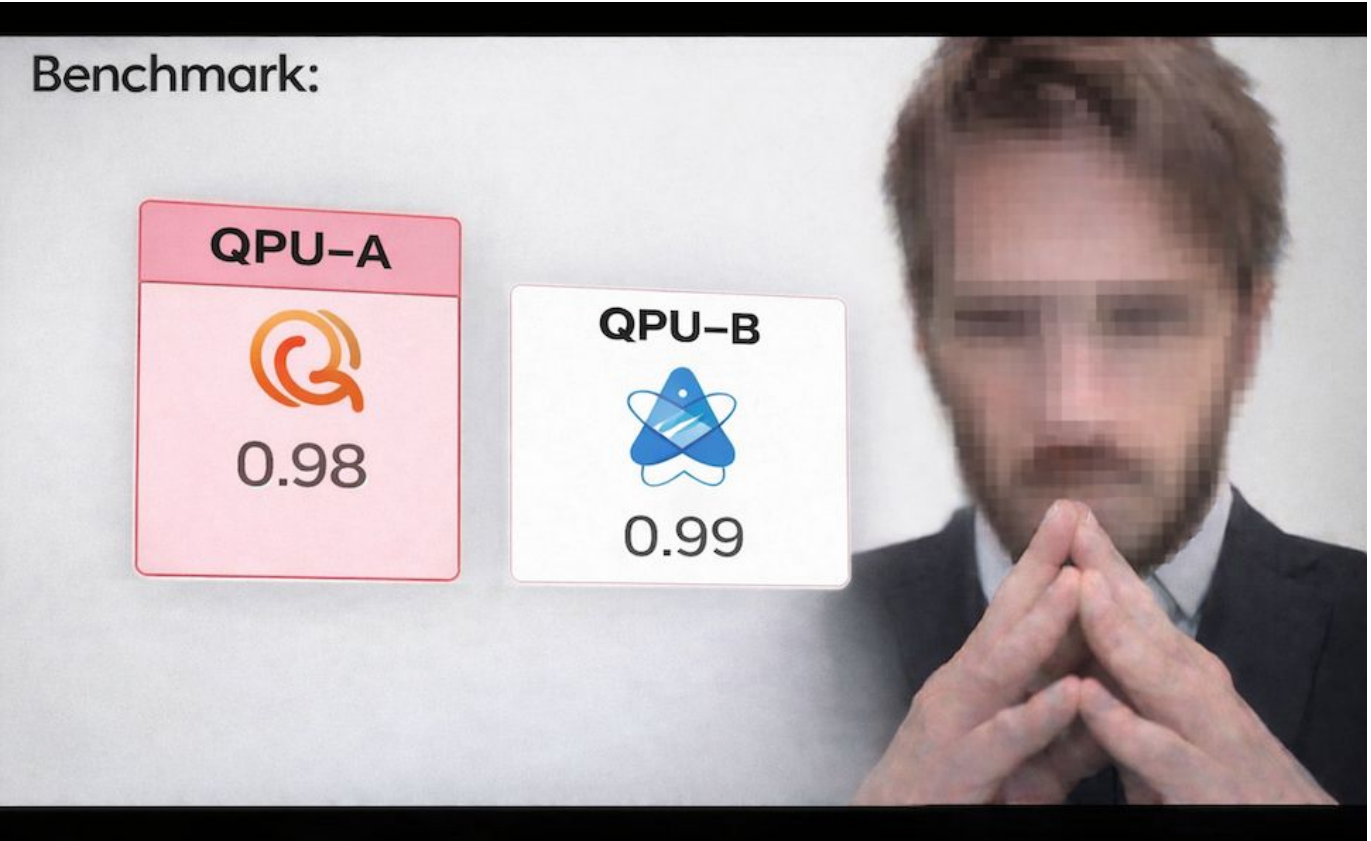
Examples (browsers)



BrowserBench · open and transparent

[from Mozilla blog, 2017]

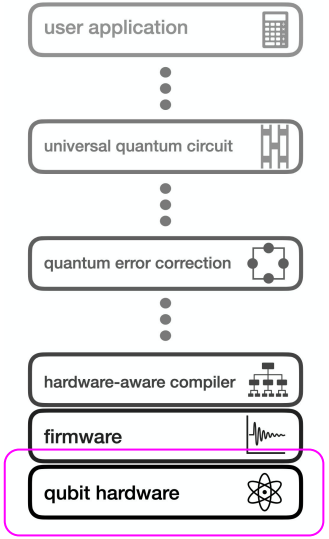
Benchmarking quantum computers



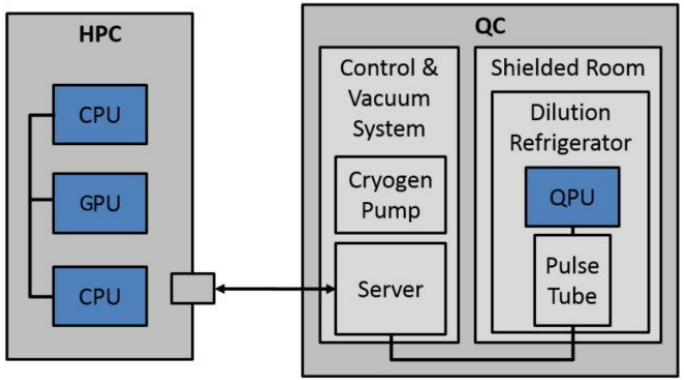
*not a real YouTube thumbnail

What do we even benchmark?

from arXiv:2501.13315

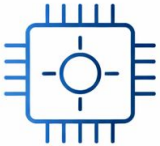


A new full-stack system



from NVIDIA blog

Multiple physical realizations of a qubit



Superconducting

Trapped ions

Neutral atoms

Photonics

Multiple physical realizations of a qubit

	Speed	Fidelity
Superconducting (IBM, Google, IQM, ...)	fast	medium
Trapped ions (Quantinuum, IonQ, ...)	slow	high
Neutral atoms (Quera, Infleqtion, ...)	medium	medium
...		

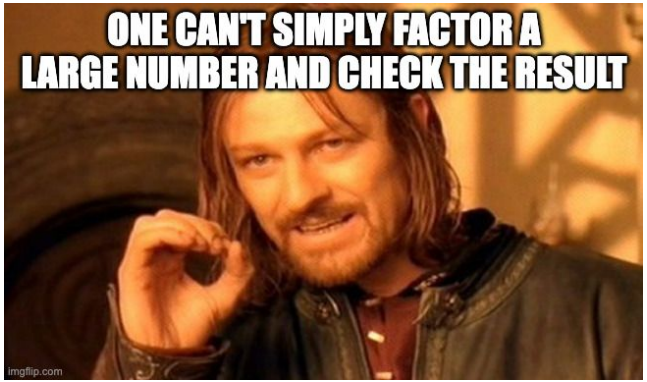
Pricing,

Scaling,

...

The NISQ (noisy intermediate-scale) reality

- Still small systems (<1000 qubits)
- Still noisy
- No application at scale yet
- Early signs of quantum advantage



Hardware is expensive 💸💸💸

	Initial Cost	Annual Operating Cost
In-House	≥ $5M	≥ $2M
Cloud Service	$0	$10K -$100K

[est. from originqc.com, 2025]

Challenges

- ❑ Multiple physical systems and architectures
- ❑ Not yet at application scale
- ❑ Access to hardware is expensive
- ❑ No standard performance metrics
- ❑ Main goal is different: make progress towards realization (for now)

The current state of play

- Vendor-specific tools
- A fragmented ecosystem
- No standardized cross-platform dataset
- Static research papers

QC-App-Oriented-Benchmarks Public

QUARK: A Framework for Quantum Computing Application Benchmarking

qiskit-device-benchmarking Public

iqm-benchmarks Public

Open-source protocols are not enough

We need open benchmarking infrastructure!

protocols · tooling · dataset · workflow

[Bonus: an open-source quantum computer!]

The Economics of an Open-Source Quantum Computer

Francesco Bova[1,2,5*] and Roger G. Melko[2,3,4,5]

4.2 As a mechanism to improve benchmarking

Utilizing an open-source project as a mechanism for benchmarking is not a new concept. For example, Imagenet, an open-source project which created a database of millions of images [23], holds a yearly contest for competing software platforms to accurately classify images from the database. This contest allows competing platforms to benchmark themselves against one another, and has led to some

Accessible tooling

Cross-platform

(run anywhere: AWS, IBM, ...)

Well-defined

(validated parameters)

Low cost

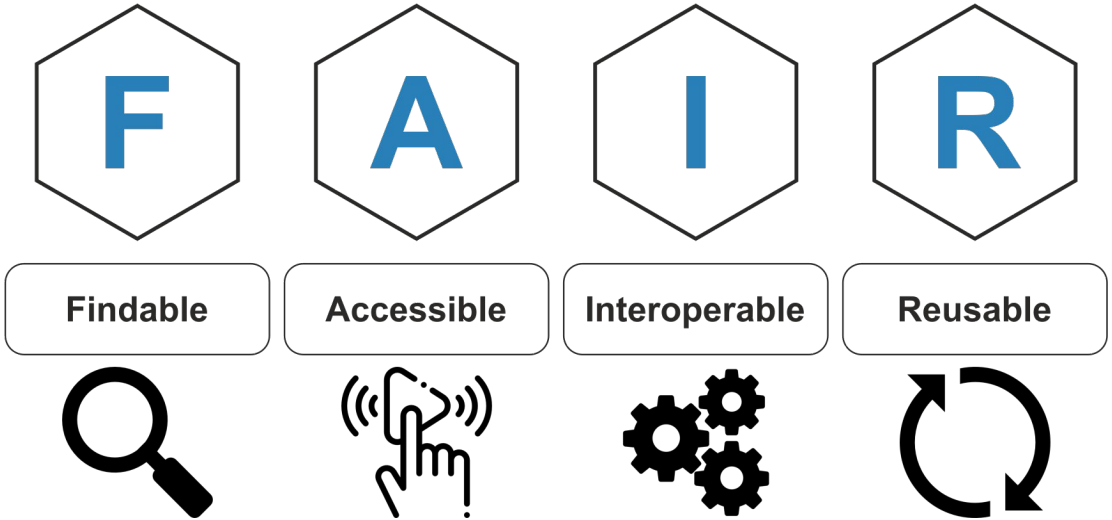
(frugal to run)

Continuously updated

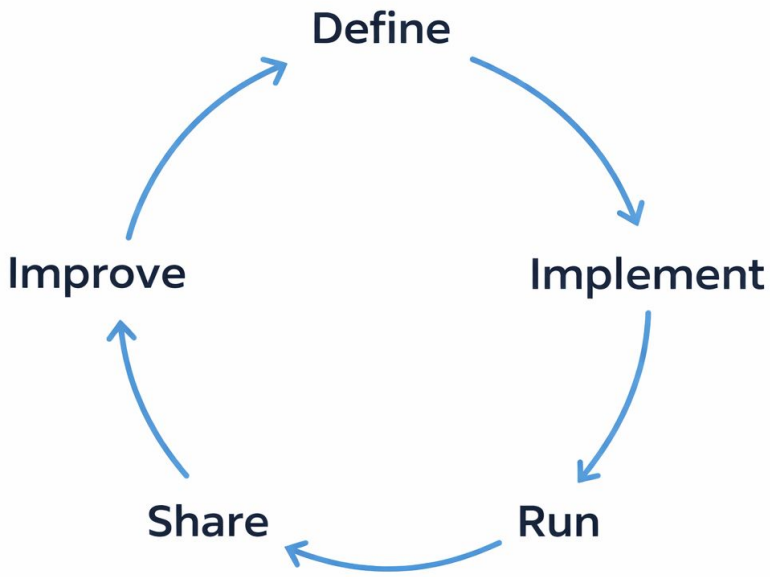
(running over time)

LF

Open cross-platform datasets



Open and accessible workflows



Metriq platform

Unitary Foundation

	Runner	Dataset	Website
/unitaryfoundation	/metriq-gym	/metriq-data	/metriq-web
License	Apache 2.0	CC-BY-4.0	Apache 2.0

Metriq Gym

```
metriq-gym

mgym suite run frugal8.json \
  --provider ibm \
  --device torino
```

- Python implementation
- Built on qBraid-SDK (Apache 2.0)
- Async jobs
  (no waiting on queues)

Metriq Suite v0.6

- System-level
- Application-inspired circuits

In Qiskit, but translatable

Metriq Data

- `Git` as a storage layer

  Out of the box

  - Version control
  - Data transparency
  - Authentication

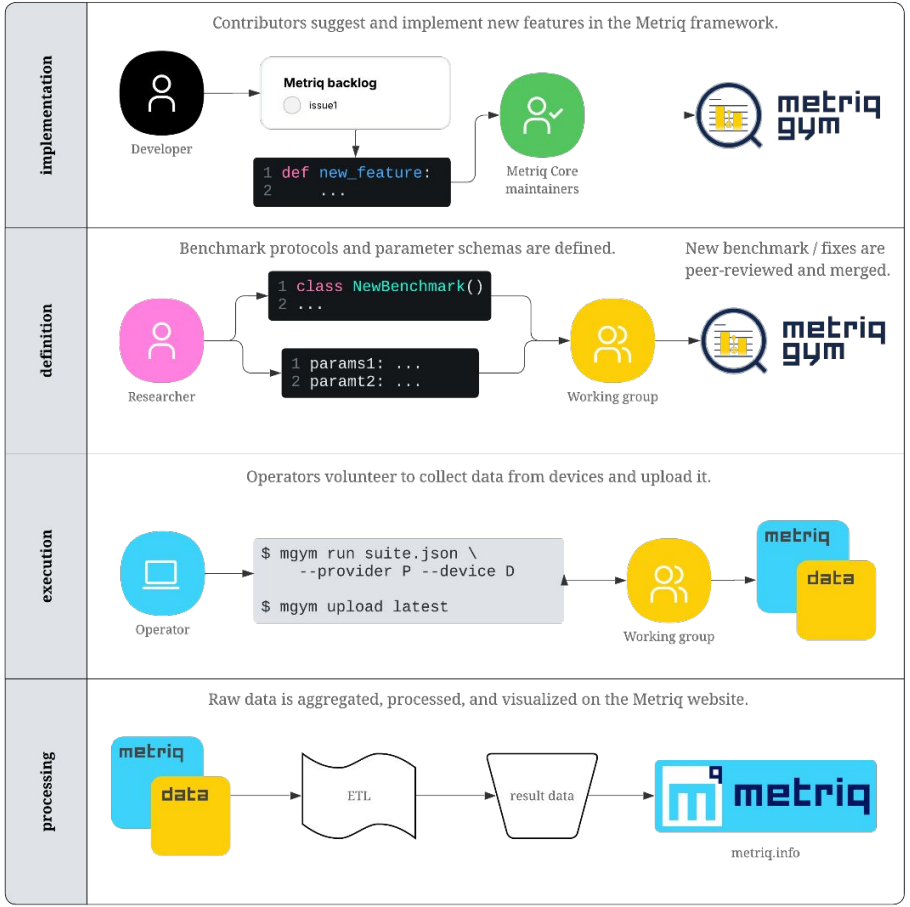
metriq-data / metriq-gym / v0.5 / ibm / ibm_boston / 2025-12-16_16-28-23_wit_35774b4e.json

cosenal mgym upload: WIT on ibm/ibm_boston (#134) caccbeb · 2 months ago History

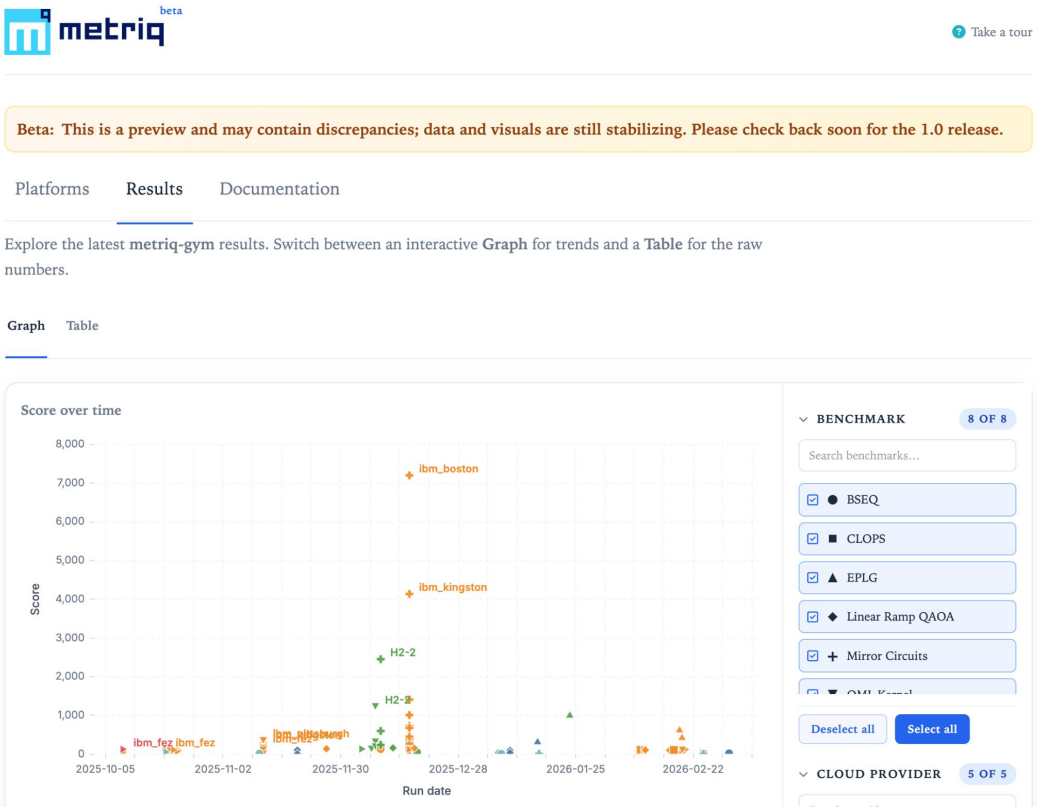
Code Blame Raw


```
[
  {
    "app_version": "0.5.1a1.dev2+g2d93ff207",
    "timestamp": "2025-12-16T16:28:23.448652",
    "suite_id": null,
    "job_type": "WIT",
    "results": {
      "expectation_value": {
        "value": 0.897216796875,
        "uncertainty": 0.00335517208526243
      },
      "score": {
        "value": 0.897216796875,
        "uncertainty": 0.00335517208526243
      }
    },
    "runtime_seconds": 6.294724,
    "platform": {
      "device": "ibm_boston",
      "device_metadata": {
        "num_qubits": 156,
        "simulator": false,
        "version": "1.0.4"
      },
      "provider": "ibm"
    },
    "params": {
      "benchmark_name": "WIT",
      "num_qubits": 7,
      "shots": 8192
    }
  }
]
```

Metriq workflow



Metriq website



Metriq Demo!

Preprint at https://arxiv.org/abs/2603.08680

new!

Metriq: A Collaborative Platform for Benchmarking Quantum Computers

Alessandro Cosentino,[1,*] Changhao Li,[1,*] Vincent Russo,[1,*] Bradley A. Chase,[1] Tom Lubinski,[2,3] Siyuan Niu,[4] Neer Patel,[4] Nathan Shammah,[1,5] and William J. Zeng[1,6]

[1] *Unitary Foundation, San Francisco, CA, USA*
[2] *Quantum Circuits, Inc., New Haven, CT, USA*
[3] *QED-C Technical Advisory Committee on Standards and Performance Benchmarks*
[4] *University of Central Florida, Orlando, FL, USA*
[5] *Unitary Fund France, Tours, France*
[6] *Quantonation, New York, NY, USA*

(Dated: March 9, 2026)

The fragmented landscape of quantum computer benchmarks, characterized by system-specific tools and inconsistent evaluation methodologies, hinders reliable cross-platform performance assessment. We introduce Metriq, an open-source collaborative platform for reproducible cross-platform quantum benchmarking that integrates benchmark definition and execution, data collection, and public presentation into a unified workflow. The Metriq benchmark suite spans both system-level metrics that characterize fundamental device properties such as entanglement quality, gate performance, and circuit speed, as well as application-inspired protocols that assess performance on quantum machine learning, optimization, and quantum simulation tasks. Benchmarks are chosen to scale with processor size, and the framework incorporates cost and resource estimation to support practical evaluation. Using Metriq, we collect and publicly release results from more than ten quantum computers across multiple hardware vendors, enabling systematic cross-platform comparison. The resulting curated dataset also reveals the practical strengths and limitations of individual benchmarks, creating a feedback loop that informs the ongoing refinement of the suite. To summarize performance across the benchmark suite, we introduce the Metriq Score, a composite index aggregating benchmark outcomes. We further present cross-benchmark analyses enabled by the shared dataset and their correlations with hardware calibration metrics. Through open development and data sharing, Metriq provides a practical foundation for reproducible benchmarking of quantum computers as hardware and benchmarking methods continue to evolve.

[quant-ph] 9 Mar 2026

What’s next

- ❑ More unified ecosystem
- ❑ More adoption
- ❑ An even more trustworthy platform
- ❑ Towards frontier benchmarks

👉 **For the community, open and accessible!**

Open quantum benchmarking needs you!

Build the tools

Design better experiences

Share benchmarking experience

Search for the **non-quantum** label

github.com/unitaryfoundation/metriq

LF

Thanks!

alessandro@unitary.foundation

cosenal.github.io

Upcoming **Talks**

Unitary Foundation × Mozilla Foundation

Mar 24
Alessandro Cosentino
Why Quantum Technology Needs Open and Accessible Benchmarking

Apr 14
Will Zeng
The Impact and Future of the Quantum Open Source Ecosystem

May 6
Ben Castanon & Veena Vijayakumar
Grantmaking and Open Source Community Development in Quantum Technology

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